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Drive circuit to drive power semiconductor element, power semiconductor module, and power conversion device

Abstract

A drive circuit for a power semiconductor element includes a voltage clamp circuit connected between a first main electrode and a control electrode and a second turn-off circuit including a capacitive element and a switching element connected in series between the control electrode and a second main electrode and a resistive element connected in parallel to the capacitive element. A turn-off control circuit switches the switching element in the second turn-off circuit to on when a voltage across the first main electrode and the second main electrode increases after start of turn-off of the power semiconductor element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on PCT filing PCT/JP2021/037882, filed Oct. 13, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a drive circuit to drive a power semiconductor element, a power semiconductor module, and a power conversion device.

BACKGROUND ART

(3) An overvoltage may occur due to a surge voltage generated when a power semiconductor element is turned off. One of measures to prevent such an overvoltage is connection of a voltage clamp circuit including a constant voltage diode between a main electrode on a high potential side and a control electrode of the power semiconductor element. When a voltage across the main electrode on the high potential side and the control electrode reaches an operating voltage level of the voltage clamp circuit, the voltage of the main electrode on the high potential side is clamped at a constant voltage and a current flows from the main electrode on the high potential side to the control electrode through the voltage clamp circuit. At this time, however, a time period for a turn-off operation becomes long due to increase in voltage of the control electrode, and hence such a side effect as increase in turn-off loss is caused.

(4) Japanese Patent Laying-Open No. 2016-86490 (PTL 1) discloses a technique to change an output-stage impedance connected to a control electrode in accordance with a value of a detected voltage of the control electrode in order to suppress increase in turn-off loss. Specifically, as surge is damped, a clamp current decreases, and consequently the voltage of the control electrode lowers. Thereafter, when a mirror period ends and when lowering in voltage of the control electrode to a value not larger than a defined value is detected, an output-stage circuit lowers the output-stage impedance.

CITATION LIST

Patent Literature

(5) PTL 1: Japanese Patent Laying-Open No. 2016-86490

SUMMARY OF INVENTION

Technical Problem

(6) In the method described in Japanese Patent Laying-Open No. 2016-86490 (PTL 1) above, however, suppression of turn-off loss may not be sufficient. Specifically, for both of a period during which the voltage across the main electrode on the high potential side and the control electrode varies and a period during which a main current that flows between main electrodes varies, the output-stage impedance is maintained at a large value. Therefore, turn-off loss is large for these periods.

(7) The present disclosure was made in consideration of problems above, and one of objects thereof is to provide a drive circuit to drive a power semiconductor element, to achieve both of suppression of a surge voltage and reduction in turn-off loss.

Solution to Problem

(8) In one embodiment, a drive circuit for a power semiconductor element is provided. The power semiconductor element includes a first main electrode on a high potential side, a second main electrode on a low potential side, and a control electrode. A state between the first main electrode and the second main electrode is switched between a conducting state and a non-conducting state in accordance with a voltage applied to the control electrode. The drive circuit includes a turn-on circuit, a first turn-off circuit, a voltage clamp circuit, a second turn-off circuit, and a turn-off control circuit. The turn-on circuit turns on the power semiconductor element by applying a first supply voltage to the control electrode in accordance with an on command from a controller. The first turn-off circuit turns off the power semiconductor element by applying a second supply voltage to the control electrode in accordance with an off command from the controller. The voltage clamp circuit is connected between the first main electrode and the control electrode. The second turn-off circuit includes a capacitive element and a switching element connected in series between the control electrode and the second main electrode and a resistive element connected in parallel to the capacitive element. The turn-off control circuit controls on and off of the switching

element in the second turn-off circuit. The turn-off control circuit switches the switching element to on when a voltage across the first main electrode and the second main electrode increases after start of turn-off of the power semiconductor element by the first turn-off circuit.

Advantageous Effects of Invention

(9) According to the drive circuit in one embodiment above, when a voltage across the first main electrode and the second main electrode increases after start of turn-off of the power semiconductor element, the switching element is switched to on. Both of suppression of the surge voltage and reduction in turn-off loss can thus be achieved.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a circuit diagram showing a configuration of a drive circuit to drive a power semiconductor element according to a first embodiment.

(2) FIG. 2 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by a drive circuit in a comparative example in which a second turn-off circuit in FIG. 1 is not provided.

(3) FIG. 3 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by the drive circuit in the first embodiment shown in FIG. 1.

(4) FIG. 4 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a second embodiment.

(5) FIG. 5 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a third embodiment.

(6) FIG. 6 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by the drive circuit in the third embodiment shown in FIG. 5.

(7) FIG. 7 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a fourth embodiment.

(8) FIG. 8 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by the drive circuit in the fourth embodiment shown in FIG. 7.

(9) FIG. 9 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a fifth embodiment.

(10) FIG. 10 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by the drive circuit in the fifth embodiment shown in FIG. 9.

(11) FIG. 11 is a circuit diagram showing a configuration of a drive circuit according to a modification of the fifth embodiment.

(12) FIG. 12 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a sixth embodiment.

(13) FIG. 13 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by the drive circuit in the sixth embodiment shown in FIG. 12.

(14) FIG. 14 is a circuit diagram showing a configuration of a drive circuit according to a modification of the sixth embodiment.

(15) FIG. 15 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a seventh embodiment.

(16) FIG. 16 is a diagram showing a voltage or current waveform of each element at the time of turn-off by the drive circuit in the seventh embodiment shown in FIG. 15.

(17) FIG. 17 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a first example of an eighth embodiment.

(18) FIG. 18 is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a second example of the eighth embodiment.

(19) FIG. **19** is a circuit diagram showing a configuration of a drive circuit to drive the power semiconductor element according to a third example of the eighth embodiment.

(20) FIG. **20** is a block diagram showing a configuration of a power conversion system to which a power conversion device according to a ninth embodiment is applied.

DESCRIPTION OF EMBODIMENTS

(21) Each embodiment will be described in detail below with reference to the drawings. In the description below, the positive logic is used for logical operation. Specifically, the high level (H level) is allocated to an active state (“1”) and the low level (L level) is allocated to an inactive state (“0”). The present disclosure, however, is applicable also to the negative logic. In the description below, the same or corresponding elements have the same reference characters allotted and description thereof may not be repeated.

First Embodiment

(22) [Exemplary Configuration of Drive Circuit]

(23) FIG. **1** is a circuit diagram showing a configuration of a drive circuit **100** to drive a power semiconductor element **1** according to a first embodiment. A power semiconductor module is composed of power semiconductor element **1** and drive circuit **100**. In this case, the power semiconductor module may be composed of power semiconductor element **1** and drive circuit **100** as being integrated or separate from each other.

(24) Power semiconductor element **1** switches a state between a first main electrode C (collector) on a high potential side and a second main electrode E (emitter) on a low potential side between a conducting (on) state and a non-conducting (off) state in accordance with a voltage applied to a control electrode G (gate). The voltage applied to control electrode G is generated by drive circuit **100** based on a command signal CMD provided from a controller **10**. Though FIG. **1** illustrates an insulated gate bipolar transistor (IGBT) as power semiconductor element **1**, the power semiconductor element is not limited to the IGBT. For example, a power semiconductor element of another type such as a power metal-oxide-semiconductor field effect transistor (MOSFET) or a reverse conducting IGBT (RC-IGBT) may be employed. A material to be used for power semiconductor element **1** is not limited to Si, and another semiconductor material such as SiC, GaN, or Ga.sub.2O.sub.3 may be employed. A freewheeling diode **2** is connected in anti-parallel to power semiconductor element **1**.

(25) Drive circuit **100** includes a turn-on circuit **11**, a first turn-off circuit **12**, a second turn-off circuit **30**, a turn-off control circuit **40**, and a voltage clamp circuit **20**.

(26) As shown in FIG. **1**, turn-on circuit **11** includes a semiconductor switching element **13** and an on gate resistor **14**. Semiconductor switching element **13** and on gate resistor **14** are connected in this order between a node (which is referred to as a VCC node below) provided with a supply voltage VCC and an intermediate node **18**. First turn-off circuit **12** includes a semiconductor switching element **16** and an off gate resistor **15**. Semiconductor switching element **16** and off gate resistor **15** are connected in this order between a node (which is referred to as a GND node below) provided with a ground potential GND and intermediate node **18**. Intermediate node **18** is connected to control electrode G of power semiconductor element **1** through a control electrode line **17**. The GND node is connected to second main electrode E of power semiconductor element **1**.

(27) In the present disclosure, more generally, supply voltage VCC may be referred to as a first supply voltage and ground potential GND may be referred to as a second supply voltage. The VCC node may be referred to as a first power supply node and the GND node may be referred to as a second power supply node.

(28) In the case of FIG. **1**, an NPN bipolar transistor is employed as semiconductor switching element **13**. A PNP bipolar transistor is employed as semiconductor switching element **16**. Without being limited as such, for example, a P-channel MOSFET may be employed as semiconductor switching element **13**, an N-channel MOSFET may be employed as semiconductor switching

element **16**, or a semiconductor switching element of another conductivity type or another type may be employed.

(29) By receiving an on command as command signal CMD from controller **10**, semiconductor switching element **13** makes transition to an on state and semiconductor switching element **16** makes transition to an off state. In the case of FIG. **1**, the on command is a voltage at the H level. Since control electrode G of power semiconductor element **1** is thus connected to the VCC node, power semiconductor element **1** is rendered conductive.

(30) By receiving an off command as command signal CMD from controller **10**, on the other hand, semiconductor switching element **13** makes transition to the off state and semiconductor switching element **16** makes transition to the on state. In the case of FIG. **1**, the off command is a voltage at the L level. Since control electrode G of power semiconductor element **1** is thus connected to the GND node, power semiconductor element **1** is rendered non-conductive.

(31) Second turn-off circuit **30** is connected between a node **34** on control electrode line **17** and second main electrode E of power semiconductor element **1**. A connection node **34** is closer to control electrode G than intermediate node **18**. As shown in FIG. **1**, by way of example, second turn-off circuit **30** includes a capacitor **31** (which is also referred to as a capacitive element), a semiconductor switching element **32**, and a resistor **33**. Capacitor **31** and semiconductor switching element **32** are connected in this order between node **34** (that is, control electrode G) and second main electrode E. Resistor **33** is connected in parallel to capacitor **31**. A passive component portion is composed of capacitor **31** and resistor **33**. While semiconductor switching element **32** is off, second turn-off circuit **30** (that is, a portion between control electrode G and second main electrode E) is rendered non-conductive, and while semiconductor switching element **32** is on, second turn-off circuit **30** is rendered conductive.

(32) In the case of FIG. **1**, an N-channel MOSFET is employed as semiconductor switching element **32**. Without being limited as such, for example, an NPN bipolar transistor or a semiconductor switching element of another conductivity type or another type may be employed as semiconductor switching element **32**.

(33) Turn-off control circuit **40** switches semiconductor switching element **32** in second turn-off circuit **30** to on when power semiconductor element **1** is turned off, that is, when a voltage (a collector-emitter voltage V_{ce}) across first main electrode C and second main electrode E increases (that is, during increase) after semiconductor switching element **13** in turn-on circuit **11** is turned off and semiconductor switching element **16** in first turn-off circuit **12** is turned on. Desirably, timing of switching of semiconductor switching element **32** is timing of start of increase in voltage across first main electrode C and second main electrode E (collector-emitter voltage V_{ce}). Thereafter, turn-off control circuit **40** sets semiconductor switching element **32** back to off after lapse of a certain time period.

(34) In the embodiment shown in FIG. **1**, turn-off control circuit **40** includes a signal generator **41A**. When signal generator **41A** receives the off command as command signal CMD from controller **10**, it outputs a pulsed signal set to the H level for a certain time period for controlling semiconductor switching element **32** to be conducting.

(35) When second turn-off circuit **30** is rendered conductive as semiconductor switching element **32** is turned on, a current flows from control electrode G of power semiconductor element **1** to the GND node through two paths including a path through first turn-off circuit **12**. A current path through first turn-off circuit **12** does not include a capacitor, whereas a current path through second turn-off circuit **30** is provided with capacitor **31** in series with semiconductor switching element **32**. Therefore, immediately after turn-on of semiconductor switching element **32**, a current instantaneously flows through second turn-off circuit **30**. The current that flows through second turn-off circuit **30** is thus significantly dominant than the current that flows through first turn-off circuit **12**. Since a turn-off operation period (that is, a period until the current that flows through second turn-off circuit **30** becomes steady) of second turn-off circuit **30** is uniquely determined by a

capacitance of capacitor **31**, it is irrelevant to a period during which a pulsed signal outputted from signal generator **41A** is at the high level. A turn-off current through second turn-off circuit **30** is thus dominant only for a very short period in accordance with the capacitance of capacitor **31**.
(36) The capacitance of capacitor **31** is set such that a transient current actually flows through second turn-off circuit **30** for a desired period. Specifically, a capacitance value of capacitor **31** is desirably set such that collector-emitter voltage V_{ce} is substantially clamped at the time when the current through capacitor **31** attains to 0.

(37) Voltage clamp circuit **20** is a circuit connected between first main electrode C on the high potential side and control electrode G of power semiconductor element **1** to clamp a voltage across first main electrode C and control electrode G. In an exemplary configuration shown in FIG. **1**, voltage clamp circuit **20** includes at least one (six) Zener diode **21** (which is also referred to as a first diode) and a backflow prevention diode **22** (which is also referred to as a second diode) connected in series with each other in this order between first main electrode C and control electrode G. Each Zener diode **21** has a cathode connected to a side of first main electrode C. Backflow prevention diode **22** has a cathode connected to a side of control electrode G. The number of Zener diodes **21** is appropriately set such that the sum of breakdown voltages of Zener diodes **21** does not exceed a rated voltage of power semiconductor element **1**. Backflow prevention diode **22** may be or may not be of a Zener type.

(38) In general, reduction in loss caused in a power semiconductor element is important for achievement of reduction in size and higher efficiency of a power conversion device including the power semiconductor element. Loss caused in the power semiconductor element includes switching loss caused in a transient state of a switching operation and conduction loss caused in the conducting state. Conduction loss is mainly determined by characteristics of the power semiconductor element, whereas switching loss can be reduced by devising a method of driving the power semiconductor element.

(39) Switching loss can be categorized into turn-on loss caused in a turn-on operation and turn-off loss caused in a turn-off operation. In order to reduce turn-off loss, a method of decreasing a turn-off time period, that is, increasing a switching speed, is available.

(40) Decrease in turn-off time period, however, brings about a problem of increase in surge voltage. Specifically, a surge voltage expressed as $L_s \times dI/dt$ is generated, where L_s represents a parasitic inductance in a main circuit of the power conversion device and dI/dt represents change over time in current (collector current) that flows between the main electrodes. Therefore, when the switching speed is increased and dI/dt becomes high, the surge voltage becomes higher. Thus, there is trade-off relation between the turn-off loss and the surge voltage.

(41) Voltage clamp circuit **20** suppresses an overvoltage involved with the surge voltage. Specifically, in the turn-off operation, the voltage across first main electrode C and second main electrode E increases. At this time, when the voltage across the main electrodes becomes excessively high due to the surge voltage, Zener diode **21** in voltage clamp circuit **20** breaks down and the current flows to control electrode G. Since power semiconductor element **1** is consequently temporarily rendered conductive, the voltage of first main electrode C on the high potential side can be maintained at a constant value. On the other hand, since a time period for the turn-off operation becomes longer due to a clamping operation by voltage clamp circuit **20**, increase in turn-off loss is disadvantageously caused. Further detailed description will be given below with reference to waveforms of a voltage and a current of each element at the time of turn-off.

(42) [Current and Voltage Waveforms in Drive Circuit in Comparative Example]

(43) FIG. **2** is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by a drive circuit in a comparative example in which second turn-off circuit **30** in FIG. **1** is not provided. FIG. **2** schematically shows sequentially from the top, a waveform of each of a voltage V_{ge} (which is also referred to as a “gate-emitter voltage”) across control electrode G and second main electrode E of power semiconductor element **1**, a main current I_c

(which is also referred to as a “collector current”) that flows between the main electrodes of power semiconductor element **1**, voltage V_{ce} (which is also referred to as the “collector-emitter voltage”) across the main electrodes of power semiconductor element **1**, and command signal CMD. A solid line indicates the waveform of the voltage or the current when voltage clamp circuit **20** performs the clamping operation and a dashed line indicates the waveform of the voltage or the current when voltage clamp circuit **20** is not provided.

(44) Referring to FIG. 2, at time t_1 , command signal CMD outputted from controller **10** in FIG. 1 switches from the H level (on command) to the L level (off command). Semiconductor switching element **13** in turn-on circuit **11** thus switches to off and semiconductor switching element **16** in first turn-off circuit **12** switches to on. Consequently, charges accumulated in control electrode G of power semiconductor element **1** start to flow out through control electrode line **17** and first turn-off circuit **12**.

(45) In the example in which voltage clamp circuit **20** is not provided (the dashed line in FIG. 2), gate-emitter voltage V_{ge} starts to lower at time t_2 , and further lowers through a mirror period from time t_3 to time t_5 . Collector-emitter voltage V_{ce} increases from time t_3 when the mirror period comes, and when the collector-emitter voltage reaches a direct-current (DC) link voltage (a voltage applied in the non-conducting state), collector current I_c starts to lower.

(46) In the example in which voltage clamp circuit **20** is provided (the solid line in FIG. 2), on the other hand, when collector-emitter voltage V_{ce} becomes higher than a clamp voltage set by voltage clamp circuit **20** at time t_4 , Zener diode **21** in voltage clamp circuit **20** breaks down and the current flows from first main electrode C (collector) to control electrode G (gate). Gate-emitter voltage V_{ge} voltage thus increases and collector-emitter voltage V_{ce} is clamped. Since collector-emitter voltage V_{ce} is thus clamped by voltage clamp circuit **20**, the surge voltage is suppressed. In the example in which voltage clamp circuit **20** is provided, however, the period for the turn-off operation of power semiconductor element **1** is longer, and hence a problem of increase in turn-off loss arises.

(47) [Current and Voltage Waveforms in Drive Circuit in First Embodiment]

(48) FIG. 3 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by the drive circuit in the first embodiment shown in FIG. 1. FIG. 3 shows sequentially from the top, a waveform of each of gate-emitter voltage V_{ge} of power semiconductor element **1**, collector current I_c , collector-emitter voltage V_{ce} , command signal CMD, and an output voltage from signal generator **41A**. The solid line indicates the waveform of the voltage or the current in the case of drive circuit **100** in the first embodiment. The dashed line indicates the waveform of the voltage or the current in the case of the comparative example in which neither of voltage clamp circuit **20** and second turn-off circuit **30** is provided in FIG. 1. Variation in waveform at time t_1 , time t_2 , time t_3 , and time t_5 is the same as in FIG. 2.

(49) At time t_{11} at which collector-emitter voltage V_{ce} starts to increase, signal generator **41A** switches the output voltage therefrom to the H level. Since semiconductor switching element **32** is thus rendered conductive, the gate current suddenly flows through capacitor **31** in second turn-off circuit **30**. Consequently, gate-emitter voltage V_{ge} lowers, collector current I_c decreases, and collector-emitter voltage V_{ce} increases.

(50) The gate current through capacitor **31** stops at next time t_{12} . A period from time t_{11} to time t_{12} is determined by the capacitance of capacitor **31**. Therefore, the capacitance value of capacitor **31** is determined in advance such that increase in collector-emitter voltage V_{ce} is also substantially clamped at time t_{12} . Thereafter, at any timing (time t_{13}) before next start of turn-on of power semiconductor element **1**, signal generator **41A** sets the output voltage therefrom back to the L level to set semiconductor switching element **32** to off.

(51) As set forth above, in the turn-off operation of power semiconductor element **1**, at least while the voltage across the main electrodes (collector-emitter voltage V_{ce}) of power semiconductor element **1** keeps increasing, signal generator **41A** maintains the output signal at the high level such

that semiconductor switching element **32** is conducting. Furthermore, the capacitance value of capacitor **31** is set such that the gate current flows through capacitor **31** only for a period during which collector-emitter voltage V_{ce} increases. In other words, the capacitance value of capacitor **31** is set such that increase in collector-emitter voltage V_{ce} is also clamped at the time when the current no longer flows through capacitor **31**.

(52) Thus, the period during which the current flows through second turn-off circuit **30** is set depending on the capacitance value of capacitor **31**, rather than the period during which semiconductor switching element **32** is conducting. It is difficult to control semiconductor switching element **32** with a pulsed signal having a pulse width as short as one hundred nanoseconds to actually allow the current to flow through second turn-off circuit **30** during the period of increase in collector-emitter voltage V_{ce} . In contrast, it is easy to determine the period during which the current flows through second turn-off circuit **30** based on the capacitance value of the capacitor.

(53) Furthermore, in second turn-off circuit **30** in FIG. 1, resistor **33** is connected in parallel to capacitor **31**. The reason therefor will be explained. When a power conversion device is assumed as an application of the power semiconductor element, the power semiconductor element is controlled to successively perform the switching operation. In other words, the turn-off operation is repeatedly performed. Therefore, in order to have the second turn-off circuit operate in every turn-off operation, charges accumulated in capacitor **31** should be released each time the turn-off operation ends. Resistor **33** is provided for discharging of capacitor **31** in such a case. In actual, when semiconductor switching element **32** is turned off, charges accumulated in capacitor **31** are discharged. In consideration of a carrier frequency and a minimum off pulse width in a pulse width modulation (PWM) signal with which the power conversion device is driven, a time constant (more generally, the time constant of the passive component portion) determined by capacitor **31** and resistor **33** is desirably not longer than five microseconds.

Effects of First Embodiment

(54) As set forth above, according to drive circuit **100** to drive power semiconductor element **1** in the first embodiment, voltage clamp circuit **20** is provided between first main electrode C on the high potential side and control electrode G of power semiconductor element **1** and second turn-off circuit **30** is provided between control electrode G and second main electrode E on the low potential side. Unlike first turn-off circuit **12**, second turn-off circuit **30** includes capacitor **31** provided in series with semiconductor switching element **32**. Furthermore, resistor **33** is connected in parallel to capacitor **31**.

(55) Turn-off control circuit **40** to control semiconductor switching element **32** in second turn-off circuit **30** turns on semiconductor switching element **32** during the turn-off period after first turn-off circuit **12** is rendered conductive. Since a current larger than the current that flows through first turn-off circuit **12** flows through capacitor **31** at this time, the voltage (collector-emitter voltage V_{ce}) across the main electrodes of power semiconductor element **1** can be increased in a shorter period of time. In other words, since variation in voltage (dV/dt) at the time of turn-off of power semiconductor element **1** becomes steeper, turn-off loss can be reduced. The capacitance value of capacitor **31** is set such that a period during which the current transiently flows through capacitor **31** is substantially equal to a period until collector-emitter voltage V_{ce} reaches the DC link voltage after semiconductor switching element **32** switches to on. Turn-off control circuit **40** turns off semiconductor switching element **32** at any timing after collector-emitter voltage V_{ce} reaches the DC link voltage. Therefore, the period during which turn-off control circuit **40** controls semiconductor switching element **32** to on is longer than a period of increase in voltage (collector-emitter voltage V_{ce}) across the main electrodes of power semiconductor element **1** from 0 to a maximum value.

(56) Since voltage clamp circuit **20** is provided in drive circuit **100**, a surge voltage involved with variation in current (dI/dt) in the turn-off operation of power semiconductor element **1** can be

suppressed. Since trade-off between the turn-off loss and the surge voltage can thus be overcome, reduction in size and higher efficiency of a power converter including power semiconductor element **1** can be achieved.

Modification of First Embodiment

(57) In drive circuit **100** in the present first embodiment, on gate resistor **14** and off gate resistor **15** are provided independently of each other. In contrast, the effect as above is obtained also when a common gate resistor for both of turn-on and turn-off is provided in control electrode line **17**.

Second Embodiment

(58) FIG. **4** is a circuit diagram showing a configuration of a drive circuit **101** to drive power semiconductor element **1** according to a second embodiment.

(59) Drive circuit **101** in the second embodiment is different from drive circuit **100** in the first embodiment in that a voltage clamp circuit **20A** further includes a capacitor **24** connected in series with a plurality of Zener diodes **21**. In an example shown in FIG. **4**, capacitor **24** is connected between first main electrode C of power semiconductor element **1** and the plurality of Zener diodes **21**. Since FIG. **4** is otherwise similar to FIG. **1**, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(60) A DC current can be cut off by providing capacitor **24**. Therefore, even when a DC voltage steadily applied across first main electrode C and second main electrode E of power semiconductor element **1** is higher than a clamp voltage determined by the number of Zener diodes **21**, the DC current does not flow through voltage clamp circuit **20**.

(61) Other effects of drive circuit **101** in the second embodiment are similar to those of drive circuit **100** in the first embodiment. Specifically, charges in control electrode G are discharged through capacitor **31** as semiconductor switching element **32** in second turn-off circuit **30** is rendered conductive during the turn-off period of power semiconductor element **1**. Consequently, since variation in voltage (dV/dt) at the time of turn-off can be steeper, turn-off loss can be reduced. Since drive circuit **101** further includes the voltage clamp circuit, the surge voltage generated across the main electrodes can be suppressed.

Third Embodiment

(62) FIG. **5** is a circuit diagram showing a configuration of a drive circuit **102** to drive power semiconductor element **1** according to a third embodiment.

(63) Drive circuit **102** in the third embodiment is different from drive circuit **101** in the second embodiment in that a second turn-off circuit **30A** further includes a negative power supply **35**. In second turn-off circuit **30A** shown in FIG. **5**, capacitor **31**, semiconductor switching element **32**, and negative power supply **35** are connected in series in this order between connection node **34** and the GND node (therefore, between control electrode G and second main electrode E). Specifically, negative power supply **35** has a positive electrode connected to the GND node and has a negative electrode connected to the main electrode (emitter) on the low potential side of semiconductor switching element **32**.

(64) According to the configuration, a potential difference between the control electrode (gate) and the main electrode (emitter) on the low potential side of semiconductor switching element **32** becomes larger. The current that flows in second turn-off circuit **30A** through semiconductor switching element **32** can thus be larger. Consequently, the rate of increase in collector-emitter voltage V_{ce} at the time of turn-off of power semiconductor element **1** can further be higher and thus the effect of reduction in turn-off loss can further be enhanced.

(65) Since FIG. **5** is otherwise the same as FIGS. **1** and **4** described above, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(66) FIG. **6** is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by drive circuit **102** in the third embodiment shown in FIG. **5**. FIG. **6** schematically shows sequentially from the top, a waveform of each of gate-emitter voltage V_{ge} of

power semiconductor element **1**, collector current I_c , collector-emitter voltage V_{ce} , and an output voltage from signal generator **41A**. The solid line indicates the waveform of the voltage or the current in the case of drive circuit **100** in the first embodiment. The dashed line indicates the waveform of the voltage or the current in the case of the comparative example in which neither of voltage clamp circuit **20** and second turn-off circuit **30** is provided in FIG. **1**. A dotted line indicates the waveform of the voltage or the current in the case of drive circuit **102** in the third embodiment. Variation in waveform at time t_2 , time t_3 , and time t_5 is the same as in FIG. **2**.

(67) As shown with the waveform drawn with the dotted line, in drive circuit **102** in the present embodiment, at time t_{14} at which collector-emitter voltage V_{ce} starts to increase, signal generator **41A** switches the output voltage therefrom to the H level. Since semiconductor switching element **32** is thus rendered conductive, a current suddenly flows through capacitor **31** in second turn-off circuit **30A**. At this time, the current that flows through second turn-off circuit **30A** is larger than in the case of second turn-off circuit **30** in the first embodiment (the solid line in FIG. **6**).

Consequently, an amount of lowering in gate-emitter voltage V_{ge} is larger than in the first embodiment, the rate of decrease in collector current I_c is higher than in the first embodiment, and the rate of increase in collector-emitter voltage V_{ce} is higher than in the first embodiment.

(68) At time t_{15} , increase in collector-emitter voltage V_{ce} is substantially clamped. The capacitance value of capacitor **31** is set such that the current that flows through capacitor **31** in second turn-off circuit **30A** also stops at this time. Thereafter, at any timing (time t_{16}) before next start of turn-on of power semiconductor element **1**, signal generator **41A** sets the output voltage therefrom back to the L level to turn off semiconductor switching element **32**. As shown in FIG. **6**, a period during which the current flows through capacitor **31** in the third embodiment is from time t_{14} until time t_{15} . This period is shorter than a period from time t_{11} until time t_{12} in the first embodiment.

(69) As set forth above, according to drive circuit **102** to drive power semiconductor element **1** in the third embodiment, negative power supply **35** is connected in series on the low potential side of semiconductor switching element **32** in second turn-off circuit **30A**. Since the current that flows through semiconductor switching element **32** can thus be increased while semiconductor switching element **32** is on, the rate of turn-off of power semiconductor element **1** can further be higher.

Consequently, variation in voltage (dV/dt) of power semiconductor element **1** can be steep and further reduction in turn-off loss can be achieved. Since drive circuit **102** includes voltage clamp circuit **20A** as in the first and second embodiments, the surge voltage transiently generated across the main electrodes of power semiconductor element **1** at the time of turn-off can be suppressed.

(70) The configuration of second turn-off circuit **30A** further including negative power supply **35** can also be combined with any of drive circuits **103** to **111** in fourth to ninth embodiments which will be described later.

Fourth Embodiment

(71) FIG. **7** is a circuit diagram showing a configuration of a drive circuit **103** to drive power semiconductor element **1** according to a fourth embodiment. Drive circuit **103** in the fourth embodiment is different from drive circuits **100** to **102** in the first to third embodiments in that a turn-off control circuit **40A** further includes a second signal generator **41B**, a delay circuit **51**, and an arithmetic logic unit **45** in addition to first signal generator **41A** described in the first to third embodiments.

(72) As described already, first signal generator **41A** outputs a pulsed signal set to the H level for a certain period based on command signal CMD (off command) outputted from controller **10**.

(73) Second signal generator **41B** detects a voltage across control electrode G and second main electrode E on the low potential side of power semiconductor element **1** and outputs a pulsed signal set to the H level for a certain period when the detected voltage becomes lower than a reference voltage V_1 . Reference voltage V_1 is set to a value smaller than the voltage of control electrode G while power semiconductor element **1** is on and larger than a mirror voltage at the time of turn-off

of power semiconductor element **1**.

(74) Delay circuit **51** delays timing of switching of output from second signal generator **41B** from the L level to the H level by predetermined delay time **DT1**. Timing of switching of semiconductor switching element **32** from off to on can thus be adjusted. When timing of turn-on of semiconductor switching element **32** is adjustable with delay time in each of a later-described comparator **44** and arithmetic logic unit **45** included in second signal generator **41B**, delay circuit **51** does not have to be provided. Delay circuit **51** may be provided between arithmetic logic unit **45** and control electrode G of semiconductor switching element **32**.

(75) When a signal obtained by delay of an output signal from second signal generator **41B** by delay circuit **51** and an output signal from first signal generator **41A** are both at the H level, arithmetic logic unit **45** outputs a signal at the H level to control electrode G of semiconductor switching element **32**. Semiconductor switching element **32** is thus rendered conductive. Arithmetic logic unit **45** thus performs a logical AND operation of a plurality of input signals.

(76) As shown in FIG. 7, more specifically, second signal generator **41B** includes a differential amplifier **42**, a DC power supply **43**, and a comparator **44**. Differential amplifier **42** functions as a control voltage detection circuit to detect a control voltage (gate-emitter voltage V_{ge}) across control electrode G and second main electrode E on the low potential side of power semiconductor element **1**. Comparator **44** compares an output voltage from differential amplifier **42** and an output voltage from DC power supply **43** with each other, and when the output voltage from differential amplifier **42** is lower than the output voltage from DC power supply **43**, it outputs a signal at the H level.

(77) Since FIG. 7 is otherwise similar to FIGS. 1 and 4, the same or corresponding elements have the same reference numerals allotted and description will not be repeated. Drive circuit **103** in the fourth embodiment can be combined with second turn-off circuit **30A** including negative power supply **35** in the third embodiment.

(78) FIG. 8 is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by drive circuit **103** in the fourth embodiment shown in FIG. 7. FIG. 8 schematically shows sequentially from the top, a waveform of each of gate-emitter voltage V_{ge} of power semiconductor element **1**, collector current I_c , collector-emitter voltage V_{ce} , an output voltage from differential amplifier **42**, an output voltage from comparator **44**, and an output voltage from arithmetic logic unit **45**. The solid line indicates the waveform of the voltage or the current in the case of drive circuit **103** in the fourth embodiment. The dashed line indicates the waveform of the voltage or the current in the case of the comparative example in which neither of voltage clamp circuit **20** and second turn-off circuit **30** is provided in FIG. 1. Variation in waveform at time t_2 , time t_3 , and time t_5 is the same as in FIG. 2.

(79) At time t_{17} in FIG. 8, the output voltage from differential amplifier **42** (that is, gate-emitter voltage V_{ge} of power semiconductor element **1**) becomes lower than reference voltage V_1 . The output voltage from comparator **44** thus switches from the L level to the H level. The output signal from first signal generator **41A** is assumed to have already switched from the L level to the H level at the time when the first signal generator receives the off command from controller **10**.

(80) At time t_{18} after lapse of delay time **DT1** by delay circuit **51** since time t_{17} , the output voltage from arithmetic logic unit **45** switches from the L level to the H level. Since semiconductor switching element **32** is thus rendered conductive, the current suddenly flows through capacitor **31** in second turn-off circuit **30**. Consequently, gate-emitter voltage V_{ge} lowers, collector current I_c decreases, and collector-emitter voltage V_{ce} increases. At next time t_{19} , the current through capacitor **31** stops. A period from time t_{18} until time t_{19} is determined by the capacitance of capacitor **31**. Therefore, the capacitance value of capacitor **31** is determined in advance such that increase in collector-emitter voltage V_{ce} is also substantially clamped at time t_{19} . Thereafter, at any timing before next start of turn-on of power semiconductor element **1**, first and second signal generators **41A** and **41B** set respective output voltages therefrom back to the L level to turn off

semiconductor switching element **32**.

Effects of Fourth Embodiment

(81) In the turn-off operation, during a mirror period during which gate-emitter voltage V_{ge} is constant, collector-emitter voltage V_{ce} increases. Therefore, as shown in FIG. **8**, reference voltage V_1 is set to a value slightly higher than a mirror voltage, and furthermore, delay time DT_1 by delay circuit **51** and a delay time produced in comparator **44** and arithmetic logic unit **45** are taken into consideration. Thus, at time t_{18} which is timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**, semiconductor switching element **32** in second turn-off circuit **30** can be switched to the conducting state. Consequently, variation in voltage (dV/dt) of power semiconductor element **1** can be steep, and further reduction in turn-off loss can be achieved. Since drive circuit **103** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage V_{ce}) of power semiconductor element **1** can be suppressed.

(82) In drive circuit **103** in the fourth embodiment, first signal generator **41A** is provided to have second turn-off circuit **30** operate at the time of turn-on of power semiconductor element **1** based on switching of command signal CMD from controller **10** from the on command to the off command. Timing of switching from off to on of semiconductor switching element **32** is determined by second signal generator **41B**.

Fifth Embodiment

(83) FIG. **9** is a circuit diagram showing a configuration of a drive circuit **104** to drive power semiconductor element **1** according to a fifth embodiment. In drive circuit **104** in the fifth embodiment, a second signal generator **41C** is different in function from second signal generator **41B** in drive circuit **103** in the fourth embodiment.

(84) Specifically, second signal generator **41C** in FIG. **9** obtains a value of a current detected by a current sensor **60** provided on a current path in first turn-off circuit **12**. Second signal generator **41C** then outputs a pulsed signal set to the H level for a certain period when an absolute value of the detected current value attains to a peak value and thereafter becomes smaller than a reference value (corresponding to a reference voltage V_2).

(85) As shown in FIG. **9**, more specifically, signal generator **41C** includes an amplifier **52**, comparator **44**, and DC power supply **43**. Amplifier **52** obtains a value of a gate current detected by current sensor **60** at the time of turn-off of power semiconductor element **1**. Amplifier **52** may be configured as a differential amplifier that obtains a differential signal from current sensor **60**. Comparator **44** compares an absolute value of the detection value from current sensor **60** outputted from amplifier **52** with reference voltage V_2 outputted from DC power supply **43**. Comparator **44** outputs the pulsed signal set to the H level for a certain period when the absolute value of the gate current detection value outputted from amplifier **52** exceeds the peak value and thereafter becomes smaller than reference voltage V_2 .

(86) Since FIG. **9** is otherwise similar in configuration to FIG. **7**, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(87) FIG. **10** is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by drive circuit **104** in the fifth embodiment shown in FIG. **9**. FIG. **10** shows sequentially from the top, a waveform of each of gate-emitter voltage V_{ge} of power semiconductor element **1**, collector current I_c , collector-emitter voltage V_{ce} , an output voltage from current sensor **60** as a gate current detector, an output voltage from comparator **44**, and an output voltage from arithmetic logic unit **45**. The solid line indicates the waveform of the voltage or the current in the case of drive circuit **104** in the fifth embodiment. The dashed line indicates the waveform of the voltage or the current in the case of the comparative example in which neither of voltage clamp circuit **20** and second turn-off circuit **30** is provided in FIG. **1**. Variation in waveform at time t_2 , time t_3 , and time t_5 is the same as in FIG. **2**. A value of the gate current is defined as being positive when the gate current flows in a direction from intermediate node **18** toward control electrode G of

power semiconductor element **1** and being negative when it flows in a direction reverse thereto.

(88) The gate current starts to flow at time **t21** in FIG. **10** and attains to the peak value at time **t22**. Thereafter, at time **t23** at which the absolute value of the gate current becomes smaller than the reference value (corresponding to reference voltage **V2**), the output voltage from comparator **44** switches from the L level to the H level. An output signal from first signal generator **41A** is assumed to have already switched from the L level to the H level when the first signal generator receives the off command from controller **10**.

(89) At time **t24** after lapse of delay time **DT2** by delay circuit **51** since time **t23**, the output voltage from arithmetic logic unit **45** switches from the L level to the H level. Since semiconductor switching element **32** is thus rendered conductive, the current suddenly flows through capacitor **31** in second turn-off circuit **30**. Consequently, gate-emitter voltage **V_{ge}** lowers, collector current **I_c** decreases, collector-emitter voltage **V_{ce}** increases, and the absolute value of the gate current increases. At next time **t25**, the current through capacitor **31** stops. A period from time **t24** until time **t25** is determined by the capacitance of capacitor **31**. Therefore, the capacitance value of capacitor **31** is determined in advance such that increase in collector-emitter voltage **V_{ce}** is also substantially clamped at time **t25**.

(90) Thereafter, at any timing before next start of turn-on of power semiconductor element **1**, first and second signal generators **41A** and **41C** set respective output voltages therefrom back to the L level to set semiconductor switching element **32** to off.

Effects of Fifth Embodiment

(91) As set forth above, according to drive circuit **104** to drive power semiconductor element **1** in the fifth embodiment, operation timing of semiconductor switching element **32** in second turn-off circuit **30** can be determined based on the detection value of the gate current that flows at the time of turn-off. Therefore, in consideration of delay time **DT2** by delay circuit **51** and the delay time produced in comparator **44** and arithmetic logic unit **45**, semiconductor switching element **32** in second turn-off circuit **30** can be switched to the conducting state at time **t24** which is timing of start of increase in collector-emitter voltage **V_{ce}** as a result of the operation of first turn-off circuit **12**. Consequently, variation in voltage (**dV/dt**) of power semiconductor element **1** can be steep and turn-off loss can further be reduced. Since drive circuit **104** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage **V_{ce}**) of power semiconductor element **1** can be suppressed.

Modification of Fifth Embodiment

(92) FIG. **11** is a circuit diagram showing a configuration of a drive circuit **105** according to a modification of the fifth embodiment. Drive circuit **105** in FIG. **11** is different from drive circuit **104** in FIG. **9** in that a second signal generator **41D** obtains a value of the gate current based on voltage lowering in off gate resistor **15** provided in first turn-off circuit **12** instead of the detection value from current sensor **60**. Therefore, differential amplifier **42** in second signal generator **41D** functions as the gate current detection circuit to detect the gate current based on a voltage across opposing ends of off gate resistor **15**.

(93) Since FIG. **11** is otherwise similar to FIGS. **7** and **9**, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

Sixth Embodiment

(94) FIG. **12** is a circuit diagram showing a configuration of a drive circuit **106** to drive power semiconductor element **1** according to a sixth embodiment. In drive circuit **106** in the sixth embodiment, a second signal generator **41E** is different in function from second signal generator **41B** in drive circuit **103** in the fourth embodiment and from second signal generators **41C** and **41D** in drive circuits **104** and **105** in the fifth embodiment.

(95) Specifically, second signal generator **41E** in FIG. **12** detects an amount of charges that flow into or out of control electrode **G** of power semiconductor element **1** by integrating the gate current detected by a gate resistor **46** provided in control electrode line **17**. Second signal generator **41E**

then outputs the pulsed signal set to the H level for a certain period when the amount of charges accumulated in control electrode G of power semiconductor element **1** at the time of turn-off becomes smaller than a reference value (corresponding to a reference voltage V4). Alternatively, signal generator **41E** may output the pulsed signal set to the H level for a certain period when the amount of charges that flow out of control electrode G at the time of turn-off becomes larger than a reference value (corresponding to a reference voltage V5 in FIG. **14**).

(96) As shown in FIG. **12**, more specifically, signal generator **41E** includes a differential amplifier **48**, an integrator **49**, comparator **44**, and DC power supply **43**. Differential amplifier **48** detects a current that flows through control electrode line **17** based on a voltage applied across opposing ends of gate resistor **46**. Integrator **49** detects the amount of charges that flow into and out of control electrode G of power semiconductor element **1** by integrating the gate current detected by differential amplifier **48**. Therefore, differential amplifier **48** and integrator **49** implement a charge amount detection circuit **47** to detect the amount of charges accumulated in control electrode G of power semiconductor element **1**. Comparator **44** outputs the pulsed signal set to the H level for a certain period when the amount of accumulated charges detected by charge amount detection circuit **47** becomes smaller than the reference value (corresponding to reference voltage V4) at the time of turn-off of power semiconductor element **1**.

(97) Since FIG. **12** is otherwise similar to FIGS. **7**, **9**, and **11**, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(98) FIG. **13** is a diagram schematically showing a voltage or current waveform of each element at the time of turn-off by drive circuit **106** in the sixth embodiment shown in FIG. **12**. FIG. **13** shows sequentially from the top, a waveform of each of gate-emitter voltage Vge of power semiconductor element **1**, collector current Ic, collector-emitter voltage Vce, an output voltage from integrator **49**, an output voltage from comparator **44**, and an output voltage from arithmetic logic unit **45**. The solid line indicates the waveform of the voltage or the current in the case of drive circuit **106** in the sixth embodiment. The dashed line indicates the waveform of the voltage or the current in the case of the comparative example in which neither of voltage clamp circuit **20** and second turn-off circuit **30** is provided in FIG. **1**. Variation in waveform at time t2, time t3, and time t5 is the same as in FIG. **2**.

(99) Gate resistor **46** in FIG. **12** can detect any of the gate current in a positive direction in flow of charges into control electrode G of power semiconductor element **1** at the time of turn-on and the gate current in a negative direction in flow of charges out of control electrode G of power semiconductor element **1** at the time of turn-off. Therefore, the output from integrator **49** monotonously increases from 0 at the time of turn-on to reach a maximum value, and monotonously decreases from the maximum value at the time of turn-off to return to 0. FIG. **13** shows as the output voltage from integrator **49**, the amount of gate charges at the time of turn-off. Therefore, after semiconductor switching element **16** in first turn-off circuit **12** switches from off to on based on the off command from controller **10** at time t1, the output voltage from integrator **49** monotonously decreases from the maximum value and returns to 0.

(100) When the output voltage from integrator **49** becomes lower than reference voltage V4 at time t30, the output voltage from comparator **44** switches from the L level to the H level. The output signal from first signal generator **41A** is assumed to have already switched from the L level to the H level when the first signal generator receives the off command from controller **10**.

(101) At time t31 after lapse of delay time DT3 by delay circuit **51** since time t30, the output voltage from arithmetic logic unit **45** switches from the L level to the H level. Since semiconductor switching element **32** is thus rendered conductive, the current suddenly flows through capacitor **31** in second turn-off circuit **30**. Consequently, gate-emitter voltage Vge lowers, collector current Ic decreases, and collector-emitter voltage Vce increases. At next time t32, the current through capacitor **31** stops. A period from time t31 until time t32 is determined by the capacitance of capacitor **31**. Therefore, the capacitance value of capacitor **31** is determined in advance such that

increase in collector-emitter voltage V_{ce} is also substantially clamped at time t_{32} .

(102) Thereafter, at any timing before next start of turn-on of power semiconductor element **1**, first and second signal generators **41A** and **41E** set respective output voltages therefrom back to the L level to turn off semiconductor switching element **32**.

(103) The output voltage from integrator **49** becomes lower than the reference voltage also in an initial stage of a turn-on operation. First signal generator **41A** is provided to prevent erroneous output during this period. First signal generator **41A** outputs the pulsed signal set to the H level for a certain time period while controller **10** outputs the off command. A final output from turn-off control circuit **40B** is obtained by logical AND of the output voltage from first signal generator **41A** and a signal obtained by delay of the output voltage from second signal generator **41E**.

Effects of Sixth Embodiment

(104) As set forth above, according to drive circuit **106** to drive power semiconductor element **1** in the sixth embodiment, operation timing of semiconductor switching element **32** in second turn-off circuit **30** can be determined based on the value of the detected amount of charges in control electrode G at the time of turn-off. Therefore, in consideration of delay time DT_3 by delay circuit **51** and the delay time produced in comparator **44** and arithmetic logic unit **45**, semiconductor switching element **32** in second turn-off circuit **30** can be switched to the conducting state at time t_{31} which is timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**. Consequently, variation in voltage (dV/dt) of power semiconductor element **1** can be steep and turn-off loss can further be reduced. Since drive circuit **106** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage V_{ce}) of power semiconductor element **1** can be suppressed.

Modification of Sixth Embodiment

(105) FIG. **14** is a circuit diagram showing a configuration of a drive circuit **107** according to a modification of the sixth embodiment. Drive circuit **107** in FIG. **14** is different from drive circuit **106** in FIG. **12** in that a second signal generator **41F** detects a value of the gate current that flows through off gate resistor **15** in first turn-off circuit **12**, instead of the value of the gate current that flows through gate resistor **46** provided in control electrode line **17**.

(106) A current flows through off gate resistor **15** in first turn-off circuit **12** in a direction from intermediate node **18** toward the GND node (in FIG. **14**, the current in this direction being defined as positive) at the time of turn-off. The amount of charges that flow out of control electrode G can be detected by integration of this current that flows through off gate resistor **15** at the time of turn-off. This amount of charges that flow out monotonously increases from 0. In order to repeatedly measure the amount of charges that flow out each time power semiconductor element **1** is turned off, an integral obtained by integrator **49** should be initialized before next start of the turn-off operation. Therefore, a charge amount detection circuit **47A** in second signal generator **41F** of drive circuit **107** includes an initialization circuit **50** to initialize the integral calculated by integrator **49** in addition to differential amplifier **48** and integrator **49**.

(107) Comparator **44** in second signal generator **41F** outputs a pulse voltage set to the H level for a certain time period when the amount of charges that flow out detected by charge amount detection circuit **47A** exceeds the reference value (corresponding to reference voltage V_5). When delay time DT_3 by delay circuit **51** has elapsed since the amount of charges that flow out exceeded the reference value, the output voltage from arithmetic logic unit **45** switches from the L level to the H level. Semiconductor switching element **32** in second turn-off circuit **30** is thus rendered conductive.

(108) As described with reference to FIGS. **12** and **13**, in consideration of delay time DT_3 by delay circuit **51** and the delay time produced in comparator **44** and arithmetic logic unit **45**, semiconductor switching element **32** in second turn-off circuit **30** can be switched to the conducting state at the timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**. Consequently, variation in voltage (dV/dt) of power

semiconductor element **1** can be steep and turn-off loss can further be reduced. Since drive circuit **107** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage V_{ce}) of power semiconductor element **1** can be suppressed.

Seventh Embodiment

(109) FIG. **15** is a circuit diagram showing a configuration of a drive circuit **108** to drive power semiconductor element **1** according to a seventh embodiment. In turn-off control circuit **40A** in drive circuit **108** in FIG. **15**, a second signal generator **41G** is different in function from second signal generators **41B**, **41C**, and **41D** in FIGS. **7**, **9**, and **11**.

(110) Specifically, second signal generator **41G** in turn-off control circuit **40A** in FIG. **15** detects a voltage applied across the anode and the cathode of backflow prevention diode **22** in voltage clamp circuit **20A**. The voltage is detected with the anode of backflow prevention diode **22** being defined as a positive side and the cathode thereof being defined as a negative side. Second signal generator **41G** outputs the pulsed signal set to the H level for a certain period when the detected voltage, the voltage having been applied across opposing ends of backflow prevention diode **22**, exceeds a reference voltage V_6 .

(111) As shown in FIG. **15**, more specifically, signal generator **41G** includes differential amplifier **42**, comparator **44**, and DC power supply **43**. Differential amplifier **42** detects the voltage applied across the anode and the cathode of backflow prevention diode **22** at the time of turn-off of power semiconductor element **1**. Comparator **44** compares the output voltage from differential amplifier **42** with reference voltage V_6 from DC power supply **43**, and when the output voltage from differential amplifier **42** exceeds reference voltage V_6 , it outputs the pulsed signal set to the H level for a certain time period.

(112) Since FIG. **15** is otherwise the same as FIGS. **7**, **9**, and **11** including second signal generators **41B**, **41C**, and **41D**, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(113) FIG. **16** is a diagram showing a voltage or current waveform of each element at the time of turn-off by drive circuit **108** in the seventh embodiment shown in FIG. **15**. FIG. **16** shows sequentially from the top, a waveform of each of gate-emitter voltage V_{ge} of power semiconductor element **1**, collector current I_c , collector-emitter voltage V_{ce} , an output voltage from differential amplifier **42**, an output voltage from comparator **44**, and an output voltage from arithmetic logic unit **45**. The solid line indicates the waveform of the voltage or the current in the case of drive circuit **108** in the seventh embodiment. The dashed line indicates the waveform of the voltage or the current in the case of the comparative example in which neither of voltage clamp circuit **20** and second turn-off circuit **30** is provided in FIG. **1**. Variation in waveform at time t_2 , time t_3 , and time t_5 is the same as in FIG. **2**.

(114) A cathode terminal of backflow prevention diode **22** is as high in potential as control electrode G of power semiconductor element **1**. Therefore, the voltage across the anode and the cathode of backflow prevention diode **22** exhibits a waveform similar to that of gate-emitter voltage V_{ge} of power semiconductor element **1**. Specifically, the voltage across the anode and the cathode of backflow prevention diode **22** has a negative value at the time of turn-off, and exhibits a waveform like gate-emitter voltage V_{ge} of power semiconductor element **1** being offset in the negative direction until start of the mirror period. Thereafter, when the mirror period comes at time t_3 and gate-emitter voltage V_{ge} attains to a constant value, in response thereto, the voltage across the anode and the cathode of backflow prevention diode **22** increases.

(115) Reference voltage V_6 of DC power supply **43** is set to a value slightly larger than the value thereof at the time when the mirror period comes. At time t_4 at which the voltage across the anode and the cathode (that is, the output voltage from differential amplifier **42**) of backflow prevention diode **22** becomes higher than reference voltage V_6 , the output signal from comparator **44** switches from the L level to the H level. An output signal from first signal generator **41A** is assumed to have already switched from the L level to the H level when the first signal generator receives the off

command from controller **10**.

(116) At time **t41** after lapse of delay time **DT4** by delay circuit **51** since time **t40**, the output voltage from arithmetic logic unit **45** switches from the L level to the H level. Since semiconductor switching element **32** is thus rendered conductive, the current suddenly flows through capacitor **31** in second turn-off circuit **30**. Consequently, gate-emitter voltage V_{ge} lowers, collector current I_c decreases, collector-emitter voltage V_{ce} increases, and the voltage across the anode and the cathode of backflow prevention diode **22** increases. At next time **t42**, the current through capacitor **31** stops. A period from time **t41** until time **t42** is determined by the capacitance of capacitor **31**. Therefore, the capacitance value of capacitor **31** is determined in advance such that increase in collector-emitter voltage V_{ce} is also substantially clamped at time **t42**.

(117) Thereafter, at any timing before next start of turn-on of power semiconductor element **1**, first and second signal generators **41A** and **41G** set the respective output voltages therefrom back to the L level to set semiconductor switching element **32** to off.

Effects of Seventh Embodiment

(118) As set forth above, according to drive circuit **108** to drive power semiconductor element **1** in the seventh embodiment, the operation timing of semiconductor switching element **32** in second turn-off circuit **30** can be determined based on the value of the detected voltage across the anode and the cathode of backflow prevention diode **22** in voltage clamp circuit **20A**. Therefore, in consideration of delay time **DT4** by delay circuit **51** and the delay time produced in comparator **44** and arithmetic logic unit **45**, semiconductor switching element **32** in second turn-off circuit **30** can be switched to the conducting state at time **t41** which is timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**. Consequently, variation in voltage (dV/dt) of power semiconductor element **1** can be steep and turn-off loss can further be reduced. Since drive circuit **108** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage V_{ce}) of power semiconductor element **1** can be suppressed.

Eighth Embodiment

(119) In an eighth embodiment, an example in which a turn-off control circuit is configured by combination of a plurality of second signal generators among second signal generators **41B** to **41G** described in the fourth to seventh embodiments will be described.

First Example

(120) FIG. **17** is a circuit diagram showing a configuration of a drive circuit **109** to drive power semiconductor element **1** according to a first example of the eighth embodiment. A turn-off control circuit **40D** of drive circuit **109** in FIG. **17** includes signal generator **41B** described with reference to FIG. **7**, signal generator **41D** described with reference to FIG. **11**, delay circuits **51A** and **51B**, and arithmetic logic unit **45**.

(121) Signal generator **41B** detects a voltage across control electrode G and second main electrode E on the low potential side of power semiconductor element **1**, and when the detected voltage becomes lower than reference voltage V_1 , the signal generator outputs the pulsed signal set to the H level for a certain period. Reference voltage V_1 is set to a value smaller than a voltage of control electrode G while power semiconductor element **1** is on and larger than a mirror voltage at the time of turn-off of power semiconductor element **1**.

(122) Delay circuit **51A** delays the output voltage from signal generator **41B** by delay time **DT1**. The output voltage from delay circuit **51A** thus switches from the L level to the H level at the timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**.

(123) Signal generator **41D** detects the gate current based on voltage lowering caused in off gate resistor **15** provided in first turn-off circuit **12**. Signal generator **41D** outputs the pulsed signal set to the H level for a certain period when the absolute value of the detected gate current exceeds the peak value and thereafter becomes smaller than a reference value (corresponding to a reference

voltage V3).

(124) Delay circuit **51B** delays the output voltage from signal generator **41D** by delay time DT2. The output voltage from delay circuit **51B** thus switches from the L level to the H level at timing of start of increase in collector-emitter voltage Vce as a result of the operation of first turn-off circuit **12**.

(125) Arithmetic logic unit **45** outputs the logical AND of outputs of delay circuit **51A** and delay circuit **51B**. As a result of switching of the output from arithmetic logic unit **45** from the L level to the H level, semiconductor switching element **32** in second turn-off circuit **30** is rendered conductive. Since FIG. **17** is otherwise the same in configuration as described in the fourth to seventh embodiments, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(126) According to the configuration above, semiconductor switching element **32** in second turn-off circuit **30** can be switched to the conducting state at timing of start of increase in collector-emitter voltage Vce as a result of the operation of first turn-off circuit **12**. Variation in voltage (dV/dt) of power semiconductor element **1** can thus be steep and turn-off loss can further be reduced. Since drive circuit **109** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage Vce) of power semiconductor element **1** can be suppressed.

Second Example

(127) FIG. **18** is a circuit diagram showing a configuration of a drive circuit **110** to drive power semiconductor element **1** according to a second example of the eighth embodiment. A turn-off control circuit **40E** of drive circuit **110** in FIG. **18** includes signal generator **41D** described with reference to FIG. **11**, signal generator **41E** described with reference to FIG. **12**, delay circuits **51A** and **51B**, and arithmetic logic unit **45**.

(128) Signal generator **41D** detects the gate current based on voltage lowering caused in off gate resistor **15** provided in first turn-off circuit **12**. Signal generator **41D** outputs the pulsed signal set to the H level for a certain period when the absolute value of the detected gate current exceeds the peak value and thereafter becomes smaller than the reference value (corresponding to reference voltage V3).

(129) Delay circuit **51A** delays the output voltage from signal generator **41D** by delay time DT2. The output voltage from delay circuit **51A** thus switches from the L level to the H level at the timing of start of increase in collector-emitter voltage Vce as a result of the operation of first turn-off circuit **12**.

(130) Signal generator **41E** detects an amount of charges that flow into and out of control electrode G of power semiconductor element **1** by integrating the gate current detected by gate resistor **46** provided in control electrode line **17**. Second signal generator **41E** then outputs the pulsed signal set to the H level for a certain period when the amount of charges accumulated in control electrode G of power semiconductor element **1** at the time of turn-off becomes smaller than the reference value (corresponding to reference voltage V4).

(131) Delay circuit **51B** delays the output voltage from signal generator **41E** by delay time DT3. The output voltage from delay circuit **51B** thus switches from the L level to the H level at timing of start of increase in collector-emitter voltage Vce as a result of the operation of first turn-off circuit **12**.

(132) Arithmetic logic unit **45** outputs the logical AND of delay circuit **51A** and delay circuit **51B**. As a result of switching of the output from arithmetic logic unit **45** from the L level to the H level, semiconductor switching element **32** in second turn-off circuit **30** is rendered conductive. Since FIG. **18** is otherwise the same in configuration as described in the fourth to seventh embodiments, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(133) According to the configuration above, semiconductor switching element **32** in second turn-

off circuit **30** can be switched to the conducting state at timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**. Variation in voltage (dV/dt) of power semiconductor element **1** can thus be steep and turn-off loss can further be reduced. Since drive circuit **110** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage V_{ce}) of power semiconductor element **1** can be suppressed.

Third Example

(134) FIG. **19** is a circuit diagram showing a configuration of a drive circuit **111** to drive power semiconductor element **1** according to a third example of the eighth embodiment. A turn-off control circuit **40F** of drive circuit **111** in FIG. **19** includes signal generator **41D** described with reference to FIG. **11**, signal generator **41G** described with reference to FIG. **15**, delay circuits **51A** and **51B**, and arithmetic logic unit **45**.

(135) Signal generator **41D** detects the gate current based on voltage lowering caused in off gate resistor **15** provided in first turn-off circuit **12**. Signal generator **41D** outputs the pulsed signal set to the H level for a certain period when the absolute value of the detected gate current exceeds the peak value and thereafter becomes smaller than the reference value (corresponding to reference voltage V_3).

(136) Delay circuit **51A** delays the output voltage from signal generator **41D** by delay time DT_2 . The output voltage from delay circuit **51A** thus switches from the L level to the H level at the timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**.

(137) Signal generator **41G** detects a voltage applied across the anode and the cathode of backflow prevention diode **22** in voltage clamp circuit **20A**. The voltage is detected with the anode of backflow prevention diode **22** being defined as the positive side and the cathode thereof being defined as the negative side. Second signal generator **41G** then outputs the pulsed signal set to the H level for a certain period when the detected voltage, the voltage having been applied across opposing ends of backflow prevention diode **22**, exceeds reference voltage V_6 .

(138) Delay circuit **51B** delays the output voltage from signal generator **41G** by delay time DT_4 . The output voltage from delay circuit **51B** thus switches from the L level to the H level at timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**.

(139) Arithmetic logic unit **45** outputs the logical AND of outputs of delay circuit **51A** and delay circuit **51B**. As a result of switching of the output from arithmetic logic unit **45** from the L level to the H level, semiconductor switching element **32** in second turn-off circuit **30** is rendered conductive. Since FIG. **19** is otherwise the same in configuration as described in the fourth to seventh embodiments, the same or corresponding elements have the same reference numerals allotted and description will not be repeated.

(140) According to the configuration above, semiconductor switching element **32** in second turn-off circuit **30** can be switched to the conducting state at timing of start of increase in collector-emitter voltage V_{ce} as a result of the operation of first turn-off circuit **12**. Variation in voltage (dV/dt) of power semiconductor element **1** can thus be steep and turn-off loss can further be reduced. Since drive circuit **111** further includes voltage clamp circuit **20A**, the voltage across the main electrodes (that is, collector-emitter voltage V_{ce}) of power semiconductor element **1** can be suppressed.

Ninth Embodiment

(141) FIG. **20** is a block diagram showing a configuration of a power conversion system to which a power conversion device according to a ninth embodiment is applied.

(142) The power conversion system shown in FIG. **20** includes a power supply **220**, a power conversion device **210**, and a load **230**. Power supply **220** is a DC power supply and supplies DC power to power conversion device **210**. Power supply **220** can be implemented by various devices

such as a DC system, a photovoltaic cell, or a storage battery, or by a rectification circuit or an alternating-current (AC)/DC converter connected to an AC system. Power supply **220** may be implemented by a DC/DC converter that converts DC power outputted from the DC system into set electric power.

(143) Power conversion device **210** is a three-phase inverter connected between power supply **220** and load **230**, and it converts DC power supplied from power supply **220** into AC power and supplies AC power to load **230**. As shown in FIG. **20**, power conversion device **210** includes a main conversion circuit **211** to convert DC power into AC power and output AC power and a control circuit **212** to output a control signal for controlling main conversion circuit **211** to main conversion circuit **211**.

(144) Load **230** is a three-phase motor driven by AC power supplied from power conversion device **210**. Load **230** is not limited to a specific application, and it is a motor mounted on various electrical appliances and used, for example, as a motor for a hybrid vehicle, an electric vehicle, a rolling stock, an elevator, or an air-conditioner.

(145) Details of power conversion device **210** will be described below. Main conversion circuit **211** includes a switching element and a freewheeling diode (not shown). As the switching element switches, DC power supplied from power supply **220** is converted into AC power, which is supplied to load **230**. Though a specific circuit configuration of main conversion circuit **211** is various, main conversion circuit **211** according to the present embodiment is a two-level three-phase full bridge circuit and can be constituted of six switching elements and six freewheeling diodes in anti-parallel to respective switching elements. At least any one of the switching elements in main conversion circuit **211** is power semiconductor element **1** in any one of the first to eighth embodiments described above. Every two switching elements of the six switching elements are connected in series to implement upper and lower arms, and the upper and lower arms implement phases (the U phase, the V phase, and the W phase) of the full bridge circuit. Output terminals of the upper and lower arms, that is, three output terminals of main conversion circuit **211**, are connected to load **230**.

(146) As described in the first to eighth embodiments above, any one (not shown) of drive circuits **100** to **111** to drive a switching element is contained in a power module **200**. Therefore, main conversion circuit **211** includes any one of drive circuits **100** to **111**. Any one of drive circuits **100** to **111** generates a drive signal to drive the switching element in main conversion circuit **211** and supplies the drive signal to the control electrode of the switching element in main conversion circuit **211**. Specifically, the drive circuit outputs to the control electrode of each switching element, a drive signal to set the switching element to on and a drive signal to set the switching element to off in accordance with a control signal from control circuit **212** which will be described later. When the switching element is to be maintained on, the drive signal is a voltage signal (on signal) not lower than a threshold voltage of the switching element, and when the switching element is to be maintained off, the drive signal is a voltage signal (off signal) not higher than a threshold voltage of the switching element.

(147) Control circuit **212** controls the switching element of main conversion circuit **211** such that desired electric power is supplied to load **230**. Control circuit **212** corresponds to controller **10** in the first to eighth embodiments. More specifically, control circuit **212** calculates a time period (an on time period) during which each switching element of main conversion circuit **211** should be on based on electric power to be supplied to load **230**. For example, main conversion circuit **211** can be controlled under PWM control in which an on time period of the switching element is modulated in accordance with a voltage to be outputted. Then, control circuit **212** outputs a control command (a control signal) to any of drive circuits **100** to **111** provided in main conversion circuit **211** such that an on signal is outputted to the switching element to be turned on and an off signal is outputted to the switching element to be turned off at each time point. Drive circuits **100** to **111** each output the on signal or the off signal as the drive signal to the control electrode of each

switching element, in accordance with the control signal.

(148) Since power semiconductor element **1** and drive circuits **101** to **111** to drive the same according to the first to eighth embodiments are applied as power module **200** included in main conversion circuit **211** in the power conversion device according to the present embodiment, a power conversion device achieving both of suppression of the surge voltage and reduction in turn-off loss can be provided.

(149) Though an example in which the present disclosure is applied to a two-level three-phase inverter is described in the present embodiment, the present disclosure can be applied to various power conversion devices without being limited as such. Though a two-level power conversion device is adopted in the present embodiment, a three-level power conversion device or a multi-level power conversion device may be applicable. In supply of electric power to a single-phase load, the present disclosure may be applied to a single-phase inverter. In supply of electric power to a DC load, the present disclosure can also be applied to a DC/DC converter or an AC/DC converter.

(150) The power conversion device to which the present disclosure is applied is not limited to an above-described example in which a motor is adopted as the load, and the power conversion device can also be employed, for example, in a power supply apparatus of an electro-discharge machine or a laser drilling machine or a power supply apparatus of an induction heating cooker or a wireless power transfer system. The power conversion device to which the present disclosure is applied can also further be used as a power conditioner of a photovoltaic power system or a power storage system.

(151) It should be understood that the embodiments disclosed herein are illustrative and non-restrictive in every respect. The scope of the present application is defined by the terms of the claims rather than the description above and is intended to include any modifications within the scope and meaning equivalent to the terms of the claims.

REFERENCE SIGNS LIST

(152) **1** power semiconductor element; **2** freewheeling diode; **10** controller; **11** turn-on circuit; **12** first turn-off circuit; **13**, **16**, **32** semiconductor switching element; **14** on gate resistor; **15** off gate resistor; **17** control electrode line; **18** intermediate node; **20**, **20A** voltage clamp circuit; **21** Zener diode (first diode); **22** backflow prevention diode (second diode); **24**, **31** capacitor; **30**, **30A** second turn-off circuit; **33** resistor; **35** negative power supply; **40**, **40A** to **40F** turn-off control circuit; **41**, **41A** to **41G** signal generator; **42**, **48** differential amplifier; **43** DC power supply; **44** comparator; **45** arithmetic logic unit; **46** gate resistor; **47**, **47A** charge amount detection circuit; **49** integrator; **50** initialization circuit; **51**, **51A**, **51B** delay circuit; **52** amplifier; **60** current sensor; **100** to **111** drive circuit; **200** power module; **210** power conversion device; **211** main conversion circuit; **212** control circuit; **220** power supply; **230** load; C first main electrode (collector); DT**1** to DT**4** delay time; E second main electrode (emitter); G control electrode (gate); V**1** to V**6** reference voltage

Claims

1. A drive circuit for a power semiconductor element, the power semiconductor element comprising a first main electrode on a high potential side, a second main electrode on a low potential side, and a control electrode, a state between the first main electrode and the second main electrode being switched between a conducting state and a non-conducting state in accordance with a voltage applied to the control electrode, the drive circuit comprising: a turn-on circuit to turn on the power semiconductor element by applying a first supply voltage to the control electrode in accordance with an on command from a controller; a first turn-off circuit to turn off the power semiconductor element by applying a second supply voltage to the control electrode in accordance with an off command from the controller; a voltage clamp circuit connected between the first main electrode and the control electrode; a second turn-off circuit comprising a capacitive element and a switching element connected in series between the control electrode and the second main electrode and a

resistive element connected in parallel to the capacitive element; and a turn-off control circuit to control on and off of the switching element in the second turn-off circuit, wherein the turn-off control circuit switches the switching element to on when a voltage across the first main electrode and the second main electrode increases after start of turn-off of the power semiconductor element by the first turn-off circuit.

2. The drive circuit according to claim 1, wherein as the turn-off control circuit turns on the switching element, for a period in accordance with a capacitance value of the capacitive element, a gate current larger than a gate current that flows through the first turn-off circuit flows through the second turn-off circuit, and the turn-off control circuit switches the switching element to off after lapse of the period in accordance with the capacitance value of the capacitive element.

3. The drive circuit according to claim 1, wherein the turn-off control circuit comprises a first signal generator, and the first signal generator receives the off command from the controller and outputs a signal for switching the switching element to on based on timing of reception of the off command.

4. The drive circuit according to claim 1, wherein the turn-off control circuit comprises a second signal generator, and the second signal generator detects a voltage across the control electrode and the second main electrode of the power semiconductor element when the power semiconductor element is turned off by the first turn-off circuit, and when the detected voltage becomes lower than a reference value, the second signal generator outputs a signal for switching the switching element to on.

5. The drive circuit according to claim 1, wherein the turn-off control circuit comprises a second signal generator, and the second signal generator detects a current that flows out of the control electrode when the power semiconductor element is turned off by the first turn-off circuit, and when an absolute value of the detected current attains to a peak value and thereafter becomes smaller than a reference value, the second signal generator outputs a signal for switching the switching element to on.

6. The drive circuit according to claim 1, wherein the turn-off control circuit comprises a second signal generator, and the second signal generator detects an amount of charges that flow out of the control electrode when the power semiconductor element is turned off by the first turn-off circuit, and when the detected amount of charges exceeds a first reference value or when an amount of charges that remain in the control electrode becomes smaller than a second reference value, the second signal generator outputs a signal for switching the switching element to on.

7. The drive circuit according to claim 1, wherein the voltage clamp circuit comprises at least one Zener type first diode for voltage clamping and a second diode connected in series in an orientation reverse to the first diode, the first diode and the second diode being connected sequentially in series between the first main electrode and the control electrode, the turn-off control circuit comprises a second signal generator, and the second signal generator detects a voltage across an anode and a cathode of the second diode when the power semiconductor element is turned off by the first turn-off circuit, and when the detected voltage exceeds a reference value, the second signal generator outputs a signal for switching the switching element to on.

8. The drive circuit according to claim 1, wherein the turn-off control circuit comprises a first signal generator, a second signal generator, and an arithmetic logic unit, the first signal generator detects a voltage across the control electrode and the second main electrode of the power semiconductor element, and when the detected voltage becomes lower than a first reference value, the first signal generator outputs a first signal, the second signal generator detects a current that flows out of the control electrode, and when an absolute value of the detected current attains to a peak value and thereafter becomes smaller than a second reference value, the second signal generator outputs a second signal, and the arithmetic logic unit outputs a signal for switching the switching element to on based on the first signal and the second signal.

9. The drive circuit according to claim 1, wherein the turn-off control circuit comprises a first

signal generator, a second signal generator, and an arithmetic logic unit, the first signal generator detects a current that flows out of the control electrode, and when an absolute value of the detected current attains to a peak value and thereafter becomes smaller than a first reference value, the first signal generator outputs a first signal, the second signal generator detects an amount of charges that flow out of the control electrode, and when the detected amount of charges exceeds a second reference value or when an amount of charges that remain in the control electrode becomes smaller than a third reference value, the second signal generator outputs a second signal, and the arithmetic logic unit outputs a signal for switching the switching element to on based on the first signal and the second signal.

10. The drive circuit according to claim 1, wherein the voltage clamp circuit comprises at least one Zener type first diode for voltage clamping and a second diode connected in series in an orientation reverse to the first diode, the first diode and the second diode being connected sequentially in series between the first main electrode and the control electrode, the turn-off control circuit comprises a first signal generator, a second signal generator, and an arithmetic logic unit, the first signal generator detects a current that flows out of the control electrode, and when an absolute value of the detected current attains to a peak value and thereafter becomes smaller than a first reference value, the first signal generator outputs a first signal, the second signal generator detects a voltage across an anode and a cathode of the second diode, and when the detected voltage exceeds a second reference value, the second signal generator outputs a second signal, and the arithmetic logic unit outputs a signal for switching the switching element to on based on the first signal and the second signal.

11. The drive circuit according to claim 1, wherein the second turn-off circuit further comprises a negative power supply, and the capacitive element, the switching element, and the negative power supply are connected in series sequentially between the control electrode and the second main electrode.

12. The drive circuit according to claim 1, wherein the second turn-off circuit comprises a passive component portion comprising the capacitive element and the resistive element, the passive component portion being connected in series with the switching element, and the passive component portion has a time constant not more than five microseconds.

13. A power semiconductor module comprising: the power semiconductor element according to claim 1; and the drive circuit according to claim 1.

14. A power conversion device incorporating: the power semiconductor element according to claim 1; and the drive circuit according to claim 1.
